COMPUTER ARCHITECTURE : Memory Technology and Hierarchy

BY

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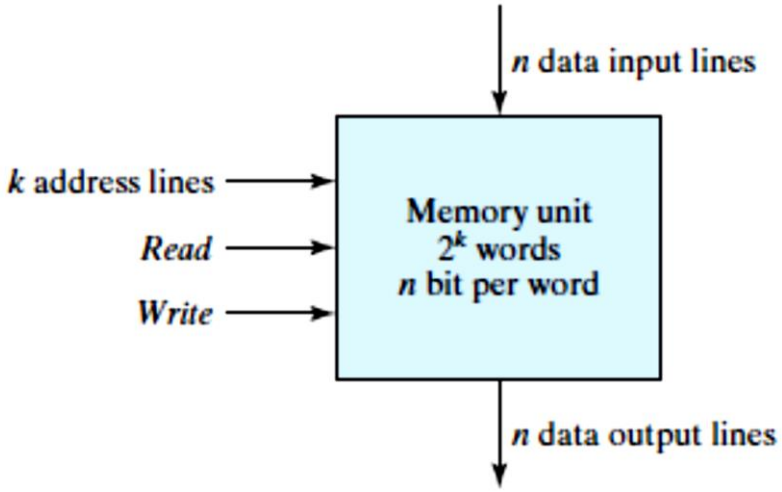
ORGANIZATION

- Memory Organization
- Read Only Memory (ROM)
- SRAM
- DRAM
- Memory hierarchy

Memory

Memory address		Memory content
Binary	Decimal	
0000000000	0	1011010101011101
0000000001	1	1010101110001001
0000000010	2	0000110101000110
⋮	⋮	⋮
1111111101	1021	1001110100010100
1111111110	1022	0000110100011110
1111111111	1023	1101111000100101

Contents of a 1024 × 16 memory



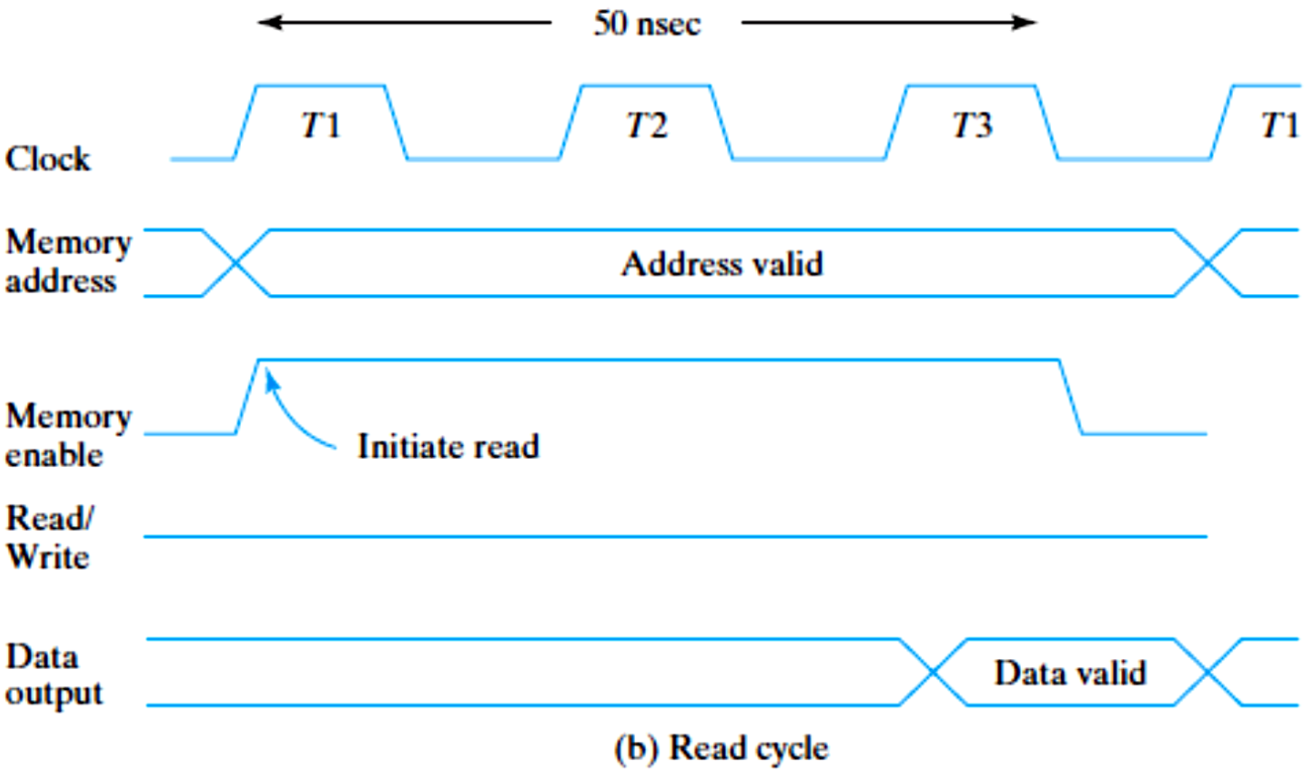
Block diagram of a memory unit

Control Inputs to Memory Chip

Memory Enable	Read/Write	Memory Operation
0	X	None
1	0	Write to selected word
1	1	Read from selected word

Memory

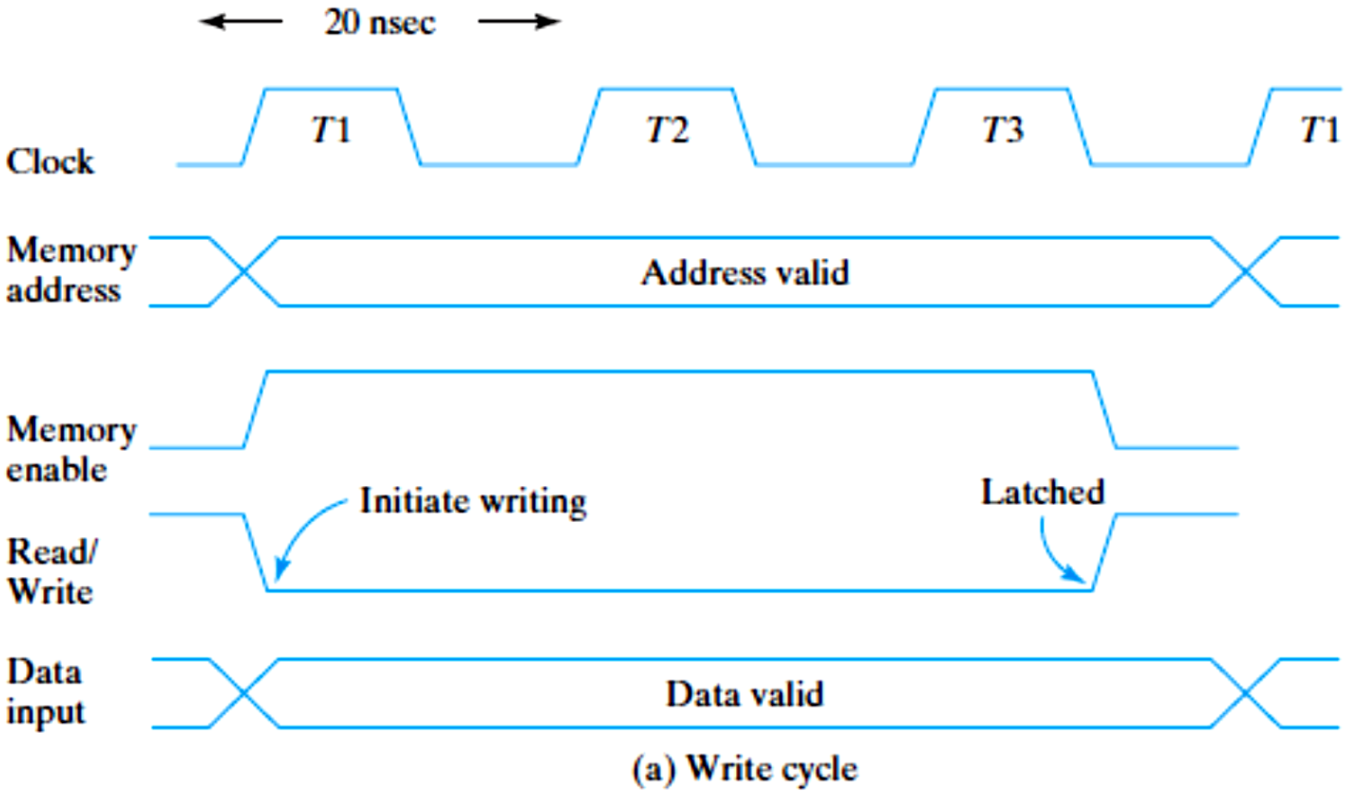
- Read cycle:



(b) Read cycle

Memory

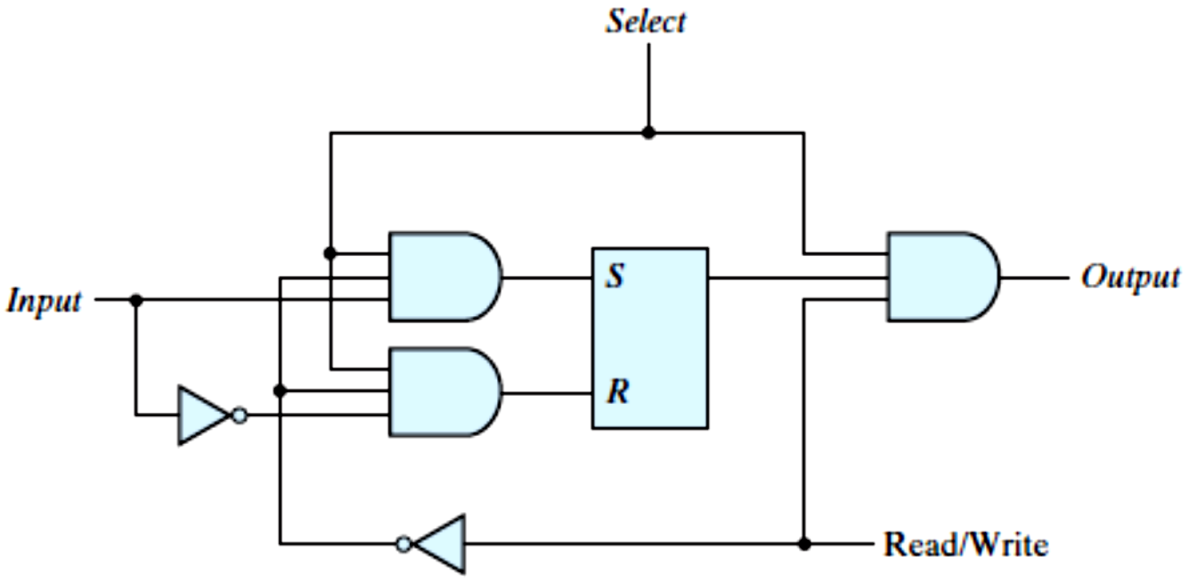
- Write cycle:



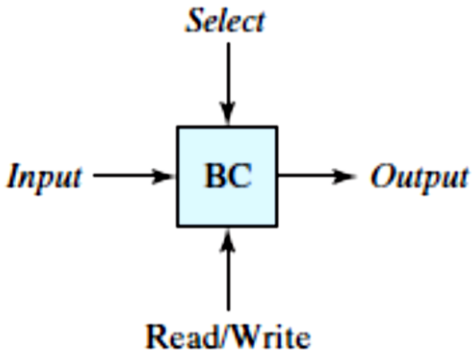
(a) Write cycle

Memory Decoding

- Memory cell:



(a) Logic diagram



(b) Block diagram

Memory

- Example:

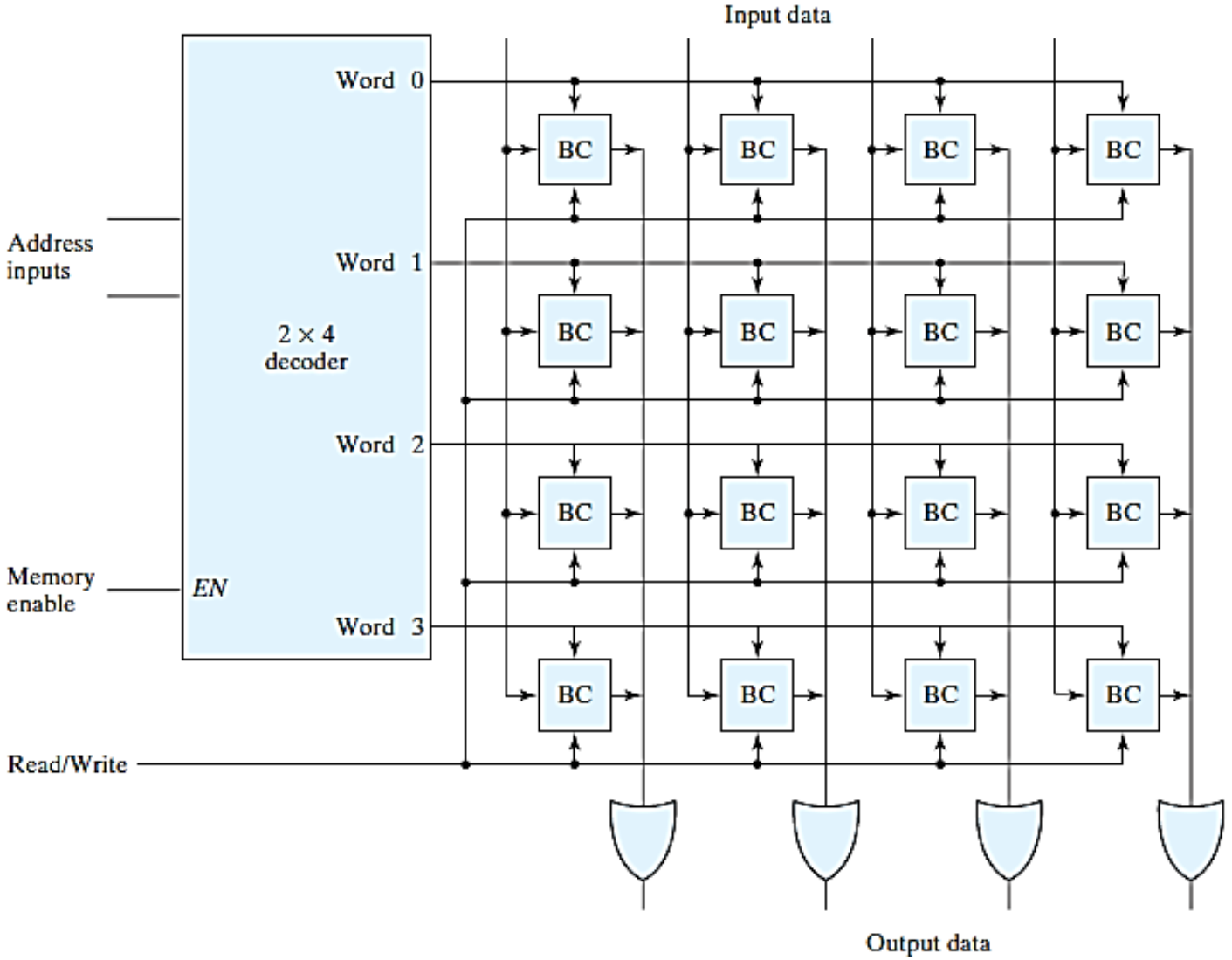
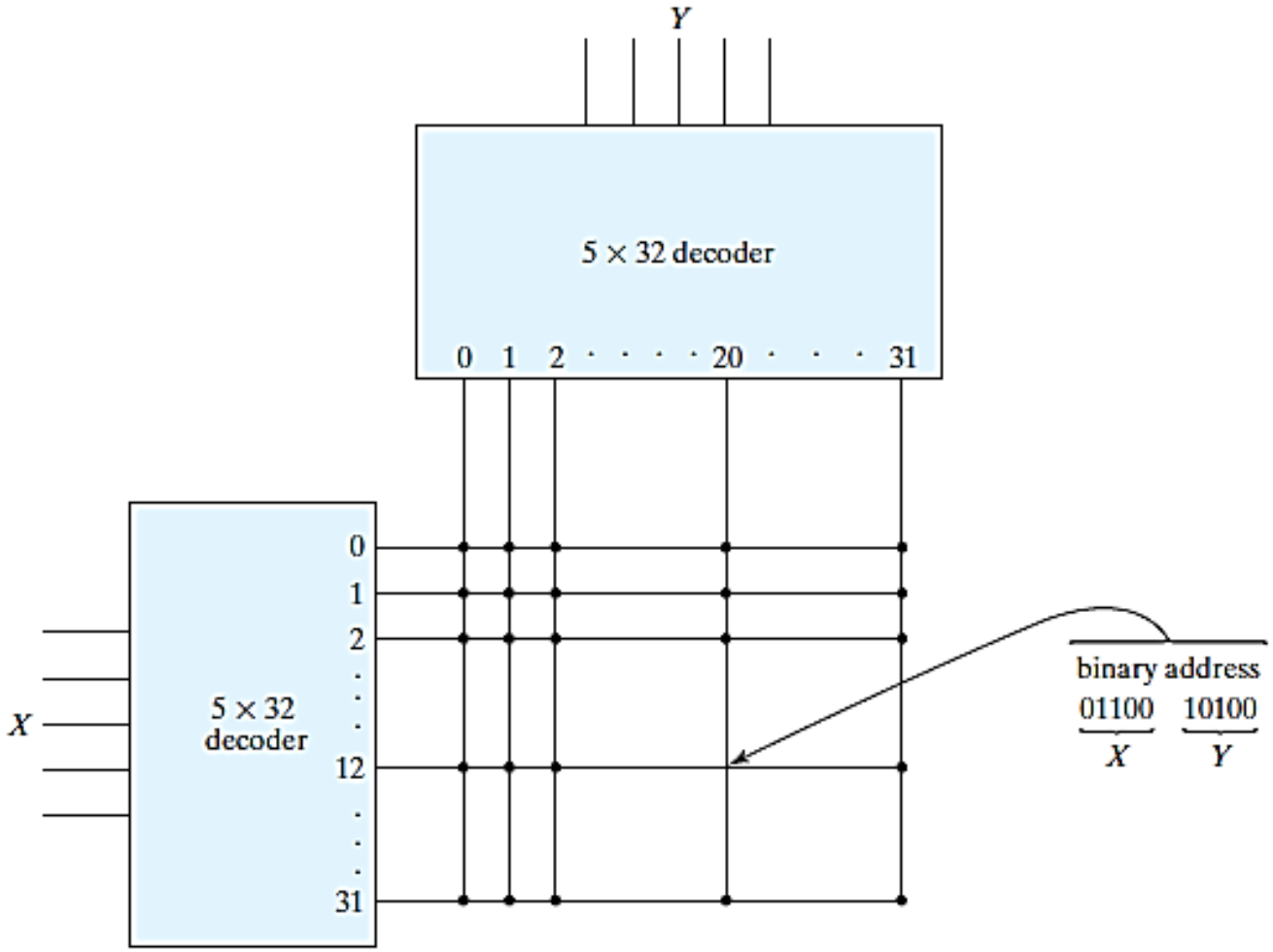


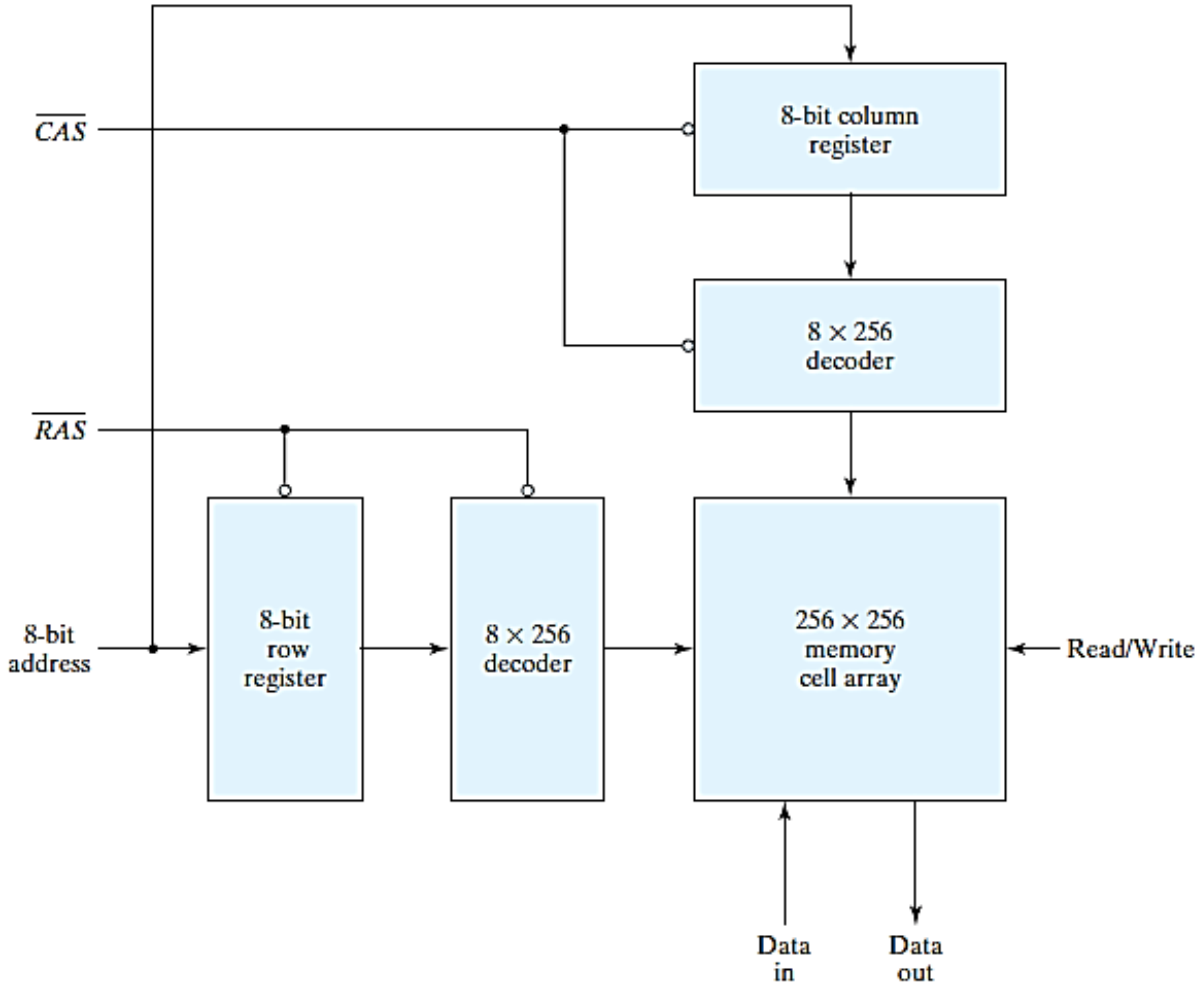
Diagram of a 4×4 RAM

Address Multiplexing



Two-dimensional decoding structure for a 1K-word memory

Address Multiplexing

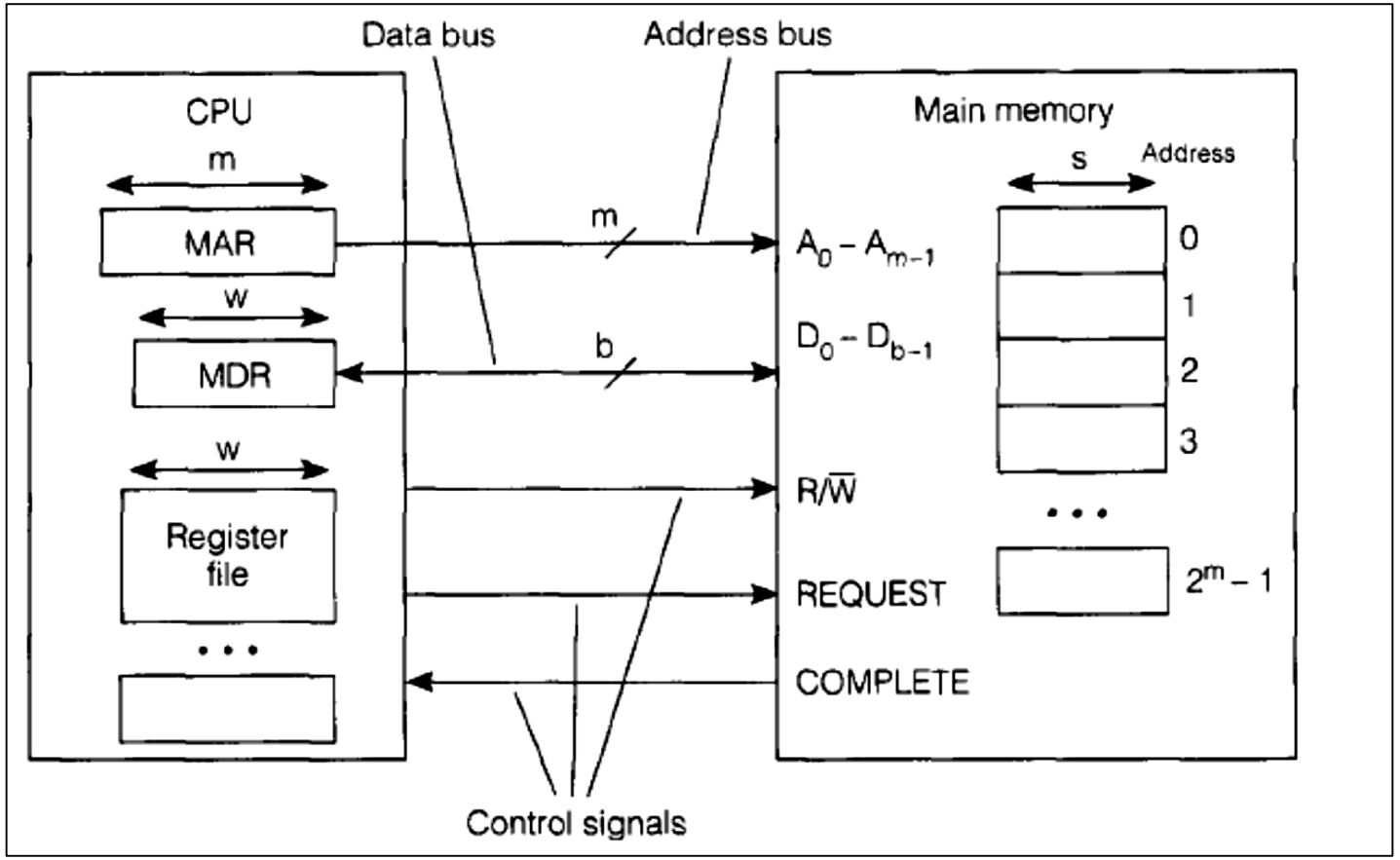


Address multiplexing for a 64K DRAM

The CPU-Memory interface

- The CPU makes a memory access request by asserting REQUEST while indicating a read or write on the R/W line.

- The memory system indicates that its process is complete by asserting COMPLETE.



Big-Endian and Little-Endian Word-Ordering Schemes

- Little-endian: Stores the least significant byte at the lower byte address and the most significant byte at the higher byte address.

- Big-endian: Stores the most significant byte at the first byte address and the least significant byte after it.

Big-Endian and Little-Endian Storage of a PowerPC 32-Bit Word

	CPU Word			
CPU word bit locations	b31 ... b24	b23 ... b16	b15 ... b8	b7 ... b0
Big-endian byte addresses	0	1	2	3
Little-endian byte addresses	3	2	1	0

Read-Only Memory (ROM)

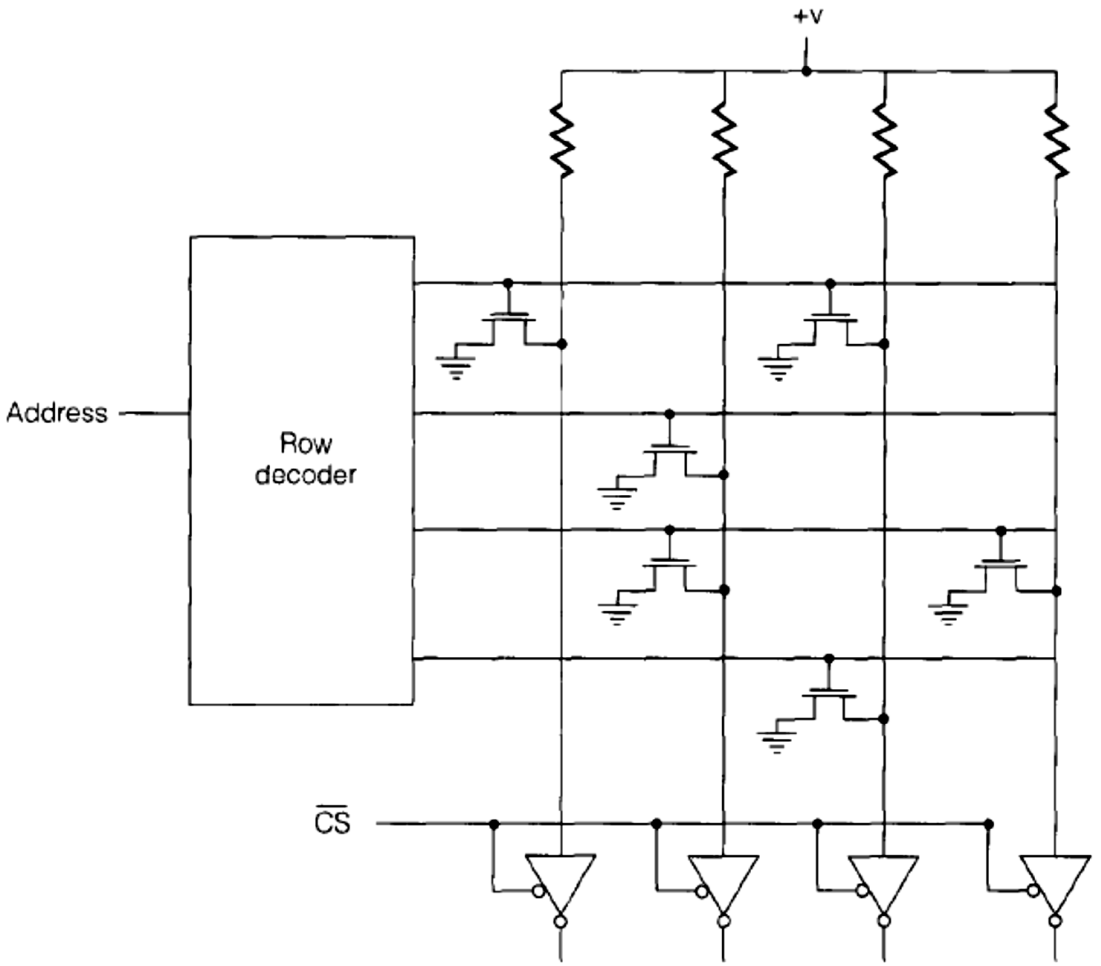
- Preprogrammed with information permanently encoded in the chip.

- ROM is by definition *nonvolatile*. That is, it retains the information in it when power is removed from it.

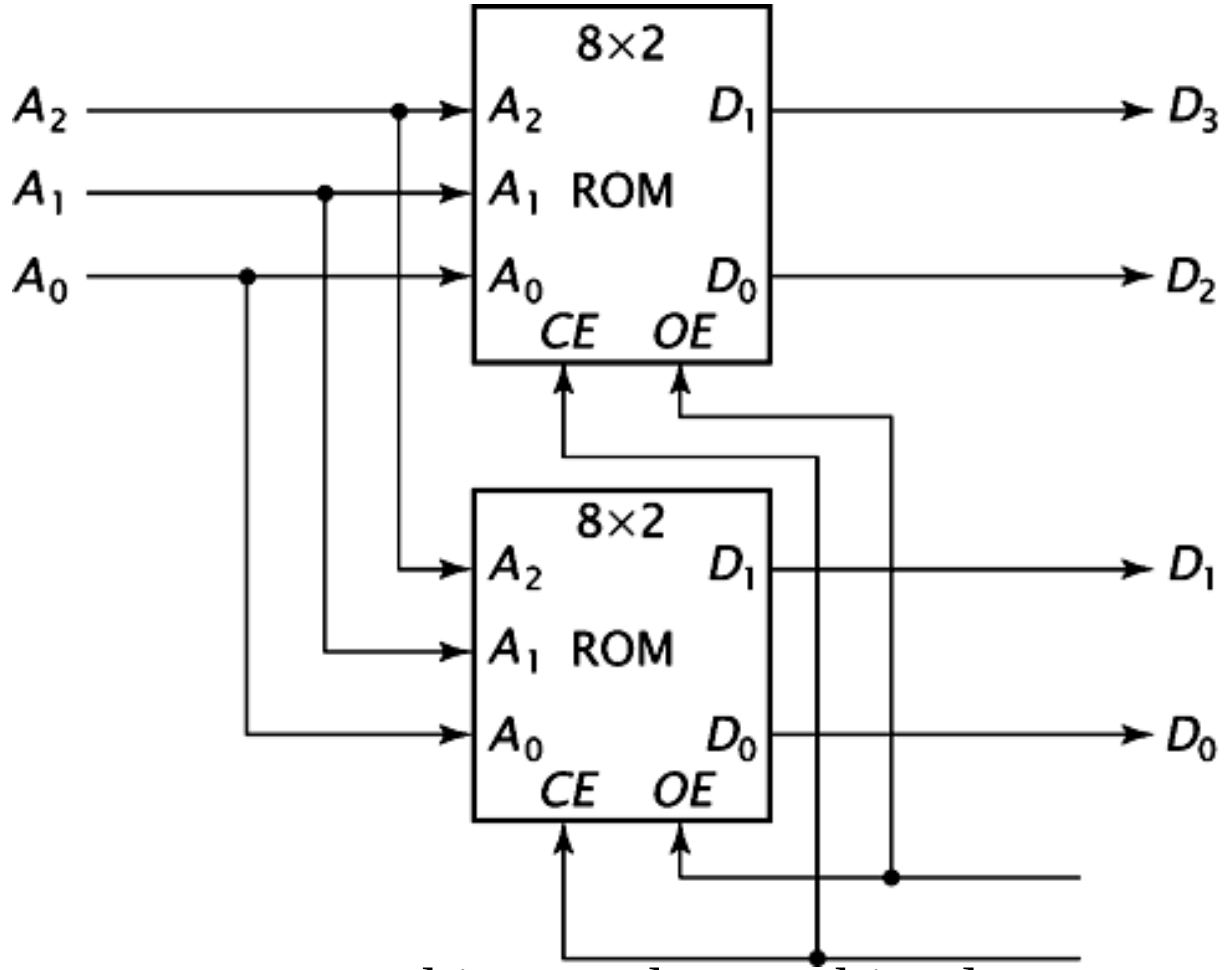
Types of ROM:

- *PROM*, programmable ROM
- *EPROM*, erasable PROM
- *EEPROM*, electrically erasable PROM

4 x 4 CMOS ROM chip

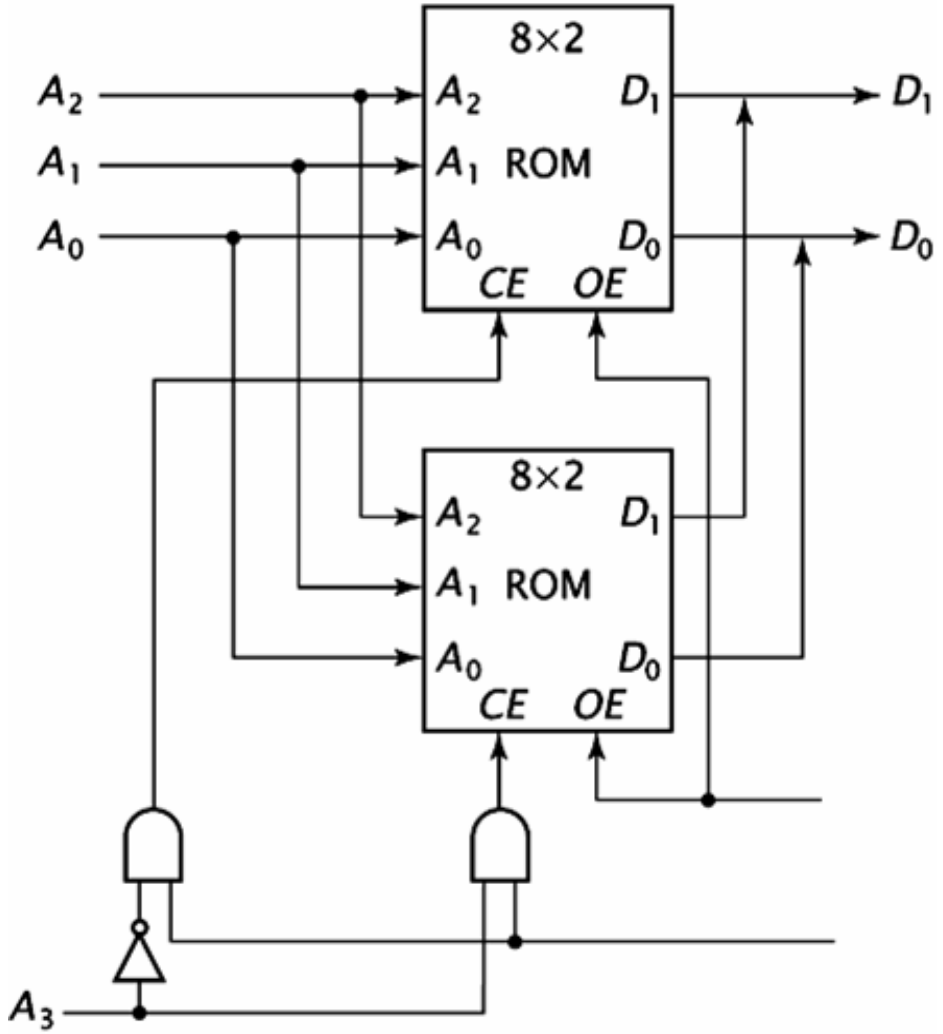


Read Only Memory (ROM)



- Two or more memory chips can be combined to create memory with more bits per location (two 8X2 chips can create a 8X4 memory)

Read Only Memory (ROM)



- Two or more memory chips can be combined to create more locations (two 8X2 chips can create 16X2 memory)

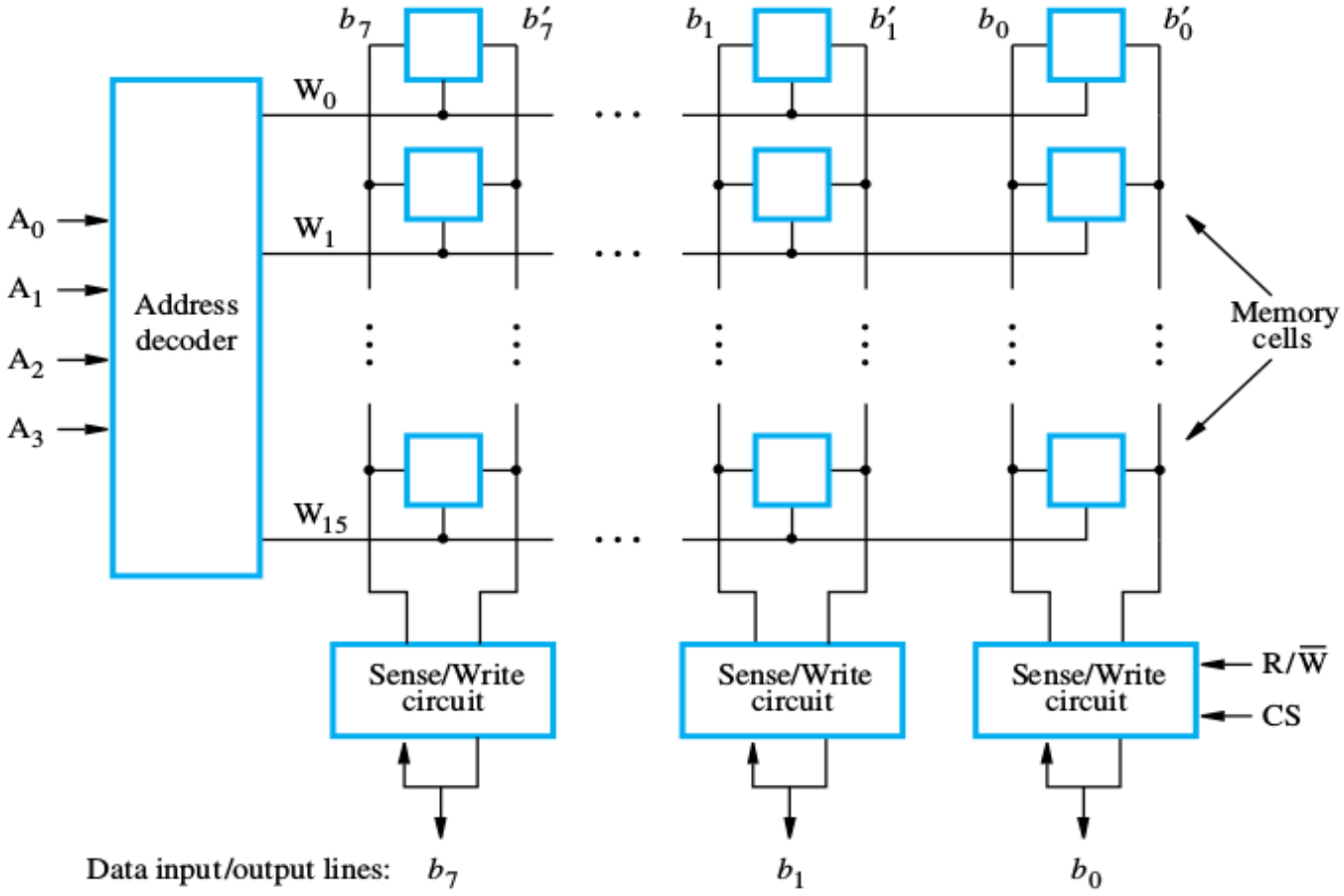
Static RAM

- Bits stored as on/off switches
- No charges to leak
- No refreshing needed when powered
- More complex construction
- Larger per bit
  - More expensive
- Does not need refresh circuits
- Faster
  - Cache
- Digital
  - Uses flip-flops

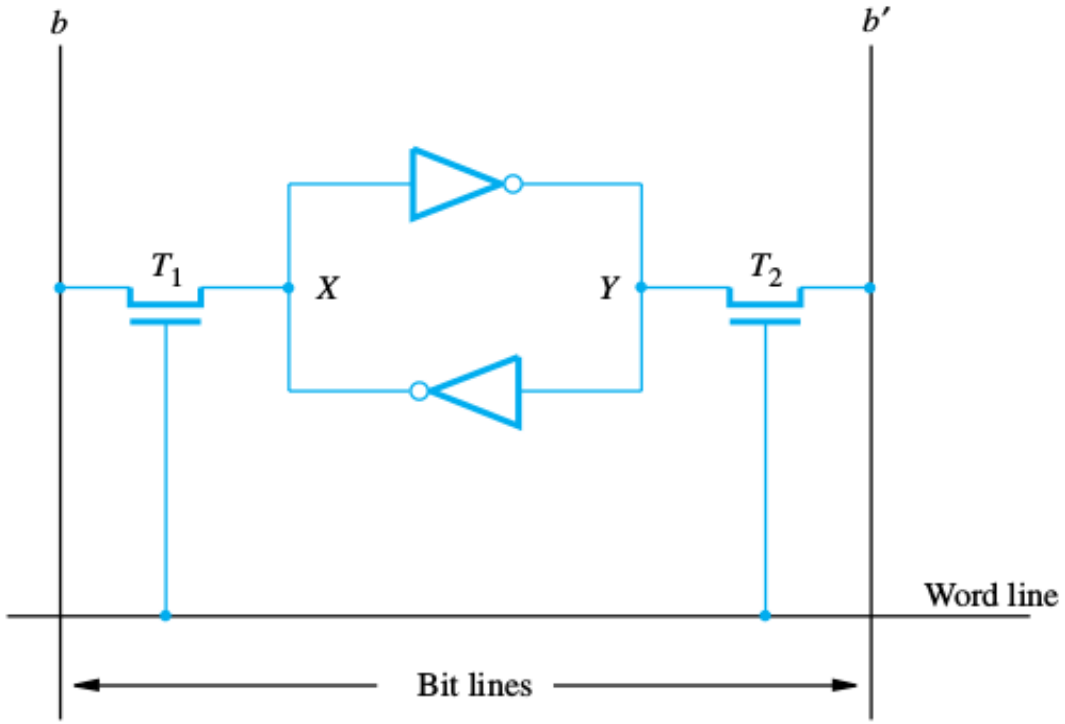
Static RAM (SRAM)

- Core cell uses six-transistor circuit to store value.
- Value is stored symmetrically—both true and complement are stored on cross-coupled transistors.
- SRAM retains value as long as power is applied to circuit.

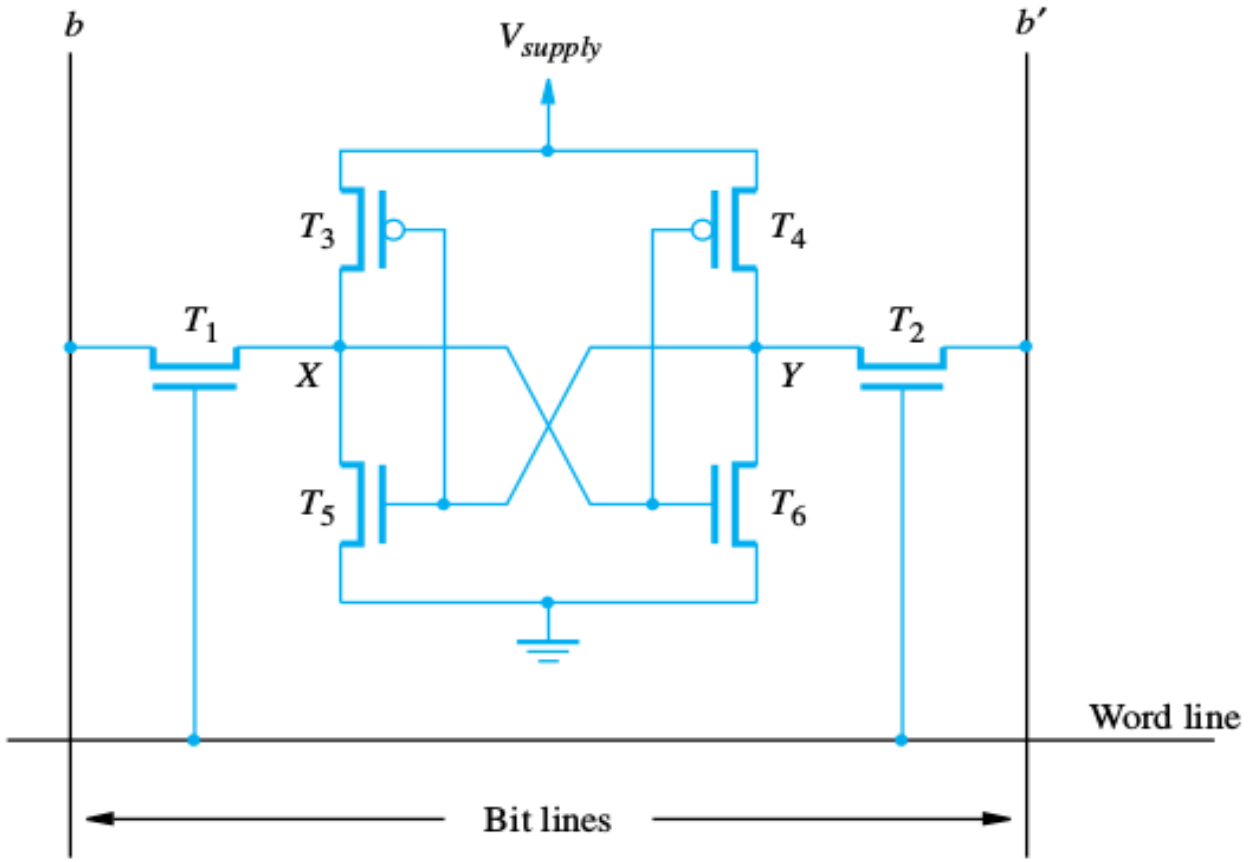
Organization of bit cells in a memory chip



SRAM cell



CMOS memory cell

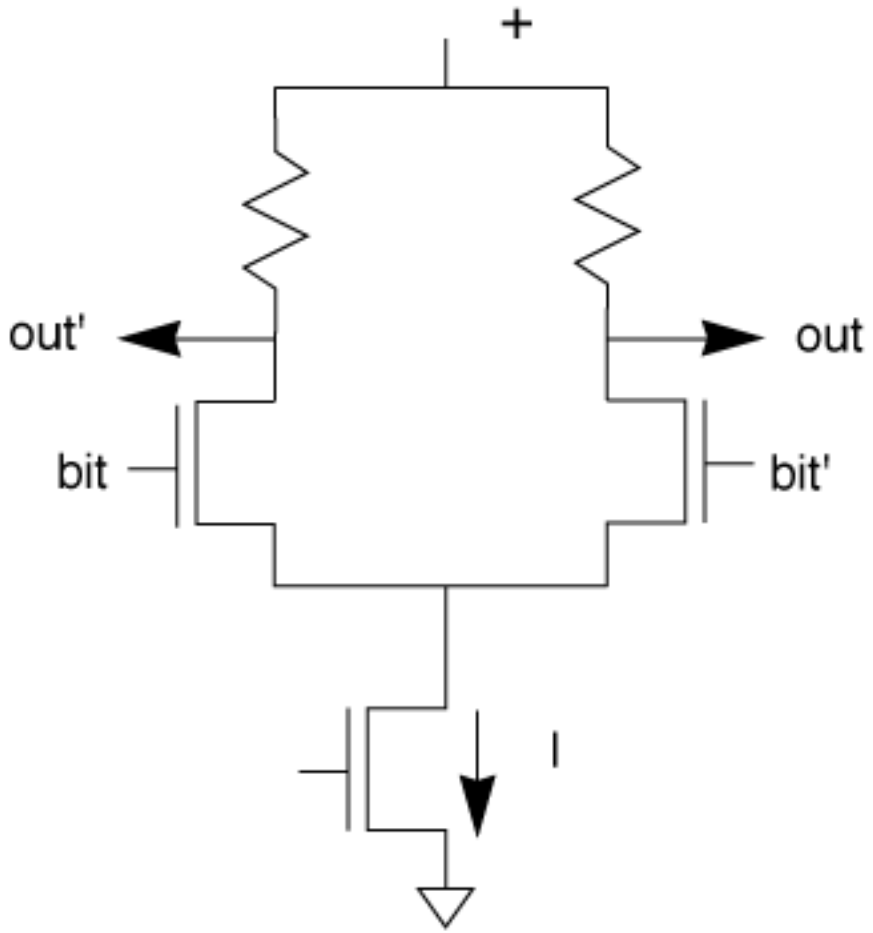


SRAM core operation

- **Read:**
  - Precharge bits b and b’
  - set word line high from row decoder; (T_1 and T_2 will be ON)
  - one bit line will be pulled down.

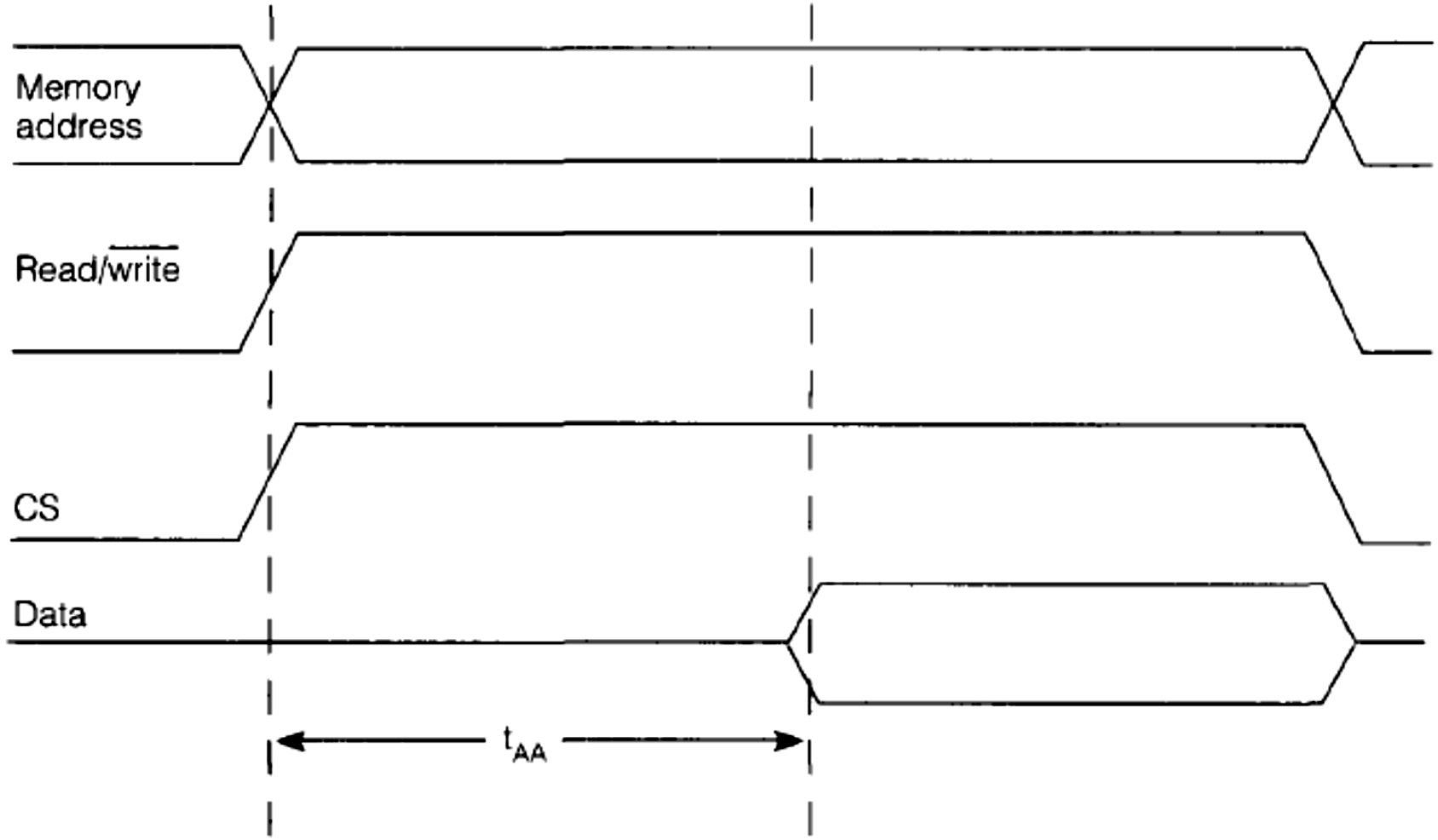
- **Write:**
  - Drive bit b and b’ to desired (complementary) values;
  - set word line high (T_1 and T_2 will be ON).
  - Drive signal given to the bit lines (b, b’) will flip the state if necessary.

SRAM sense amplifier

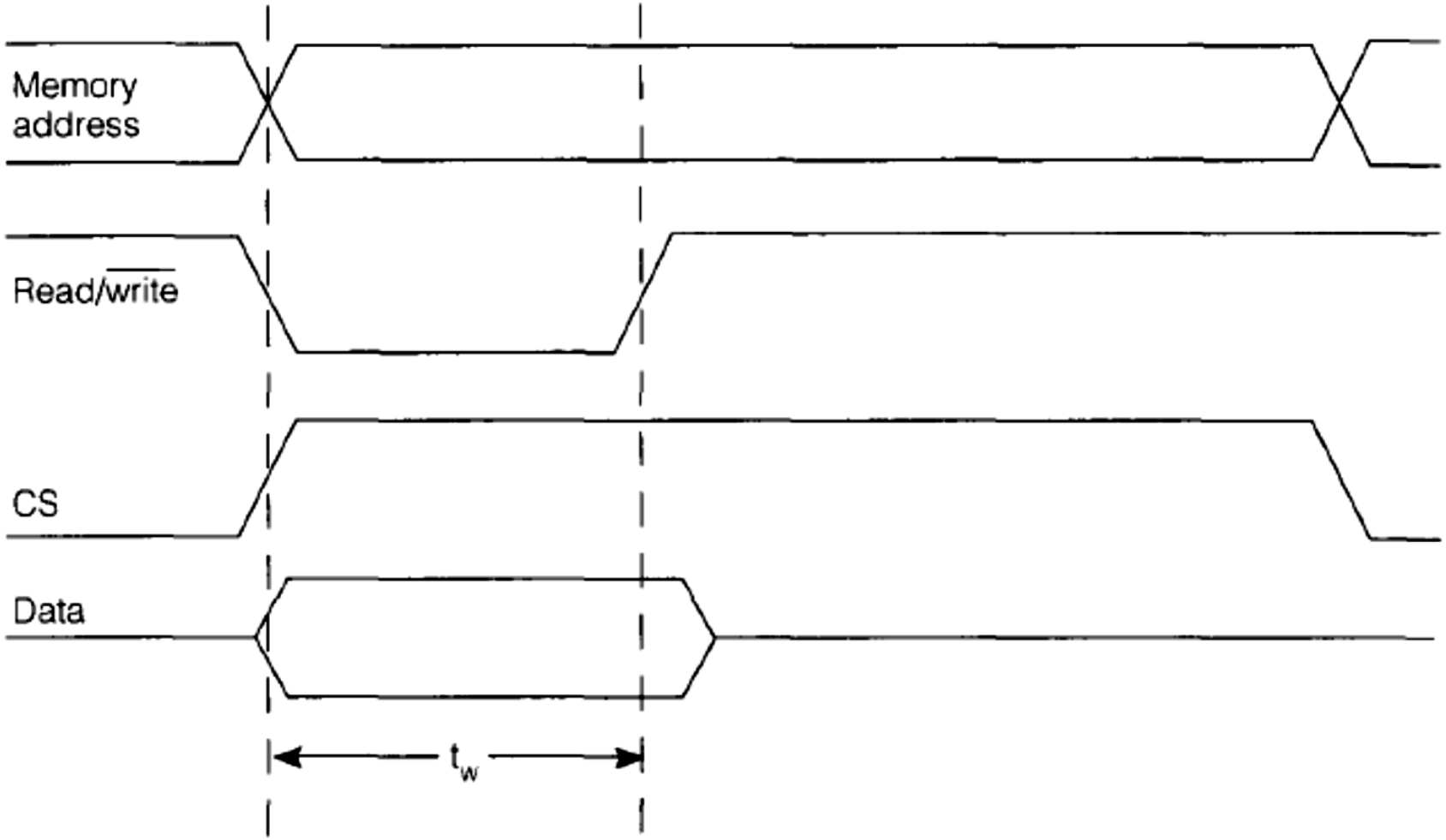


- Differential pair—takes advantage of complementarity of bit lines.
- When one bit line goes low, that arm of diff pair reduces its current, causing compensating increase in current in other arm.
- Sense amp can be cross-coupled to increase speed.

SRAM Timing: Read operation



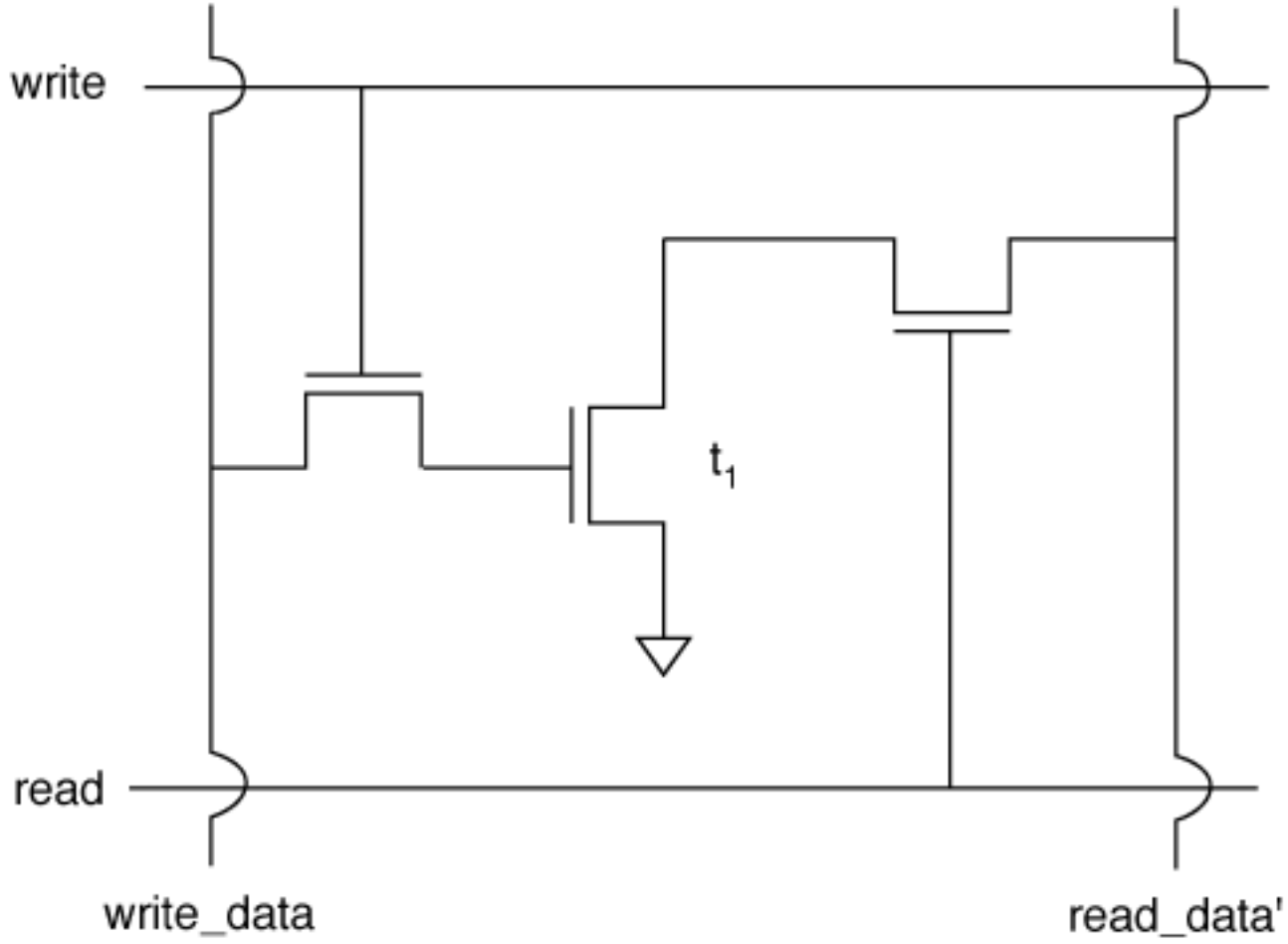
SRAM Timing: Write operation



Dynamic RAM

- Bits stored as charge in capacitors
  - Charges leak
  - Need refreshing even when powered
- Simpler construction
- Smaller per bit than SRAM
  - Less expensive
- Need refresh circuits
- Slower
- Used for main memory in computing systems
- Essentially analogue
  - Level of charge determines value

3-Transistor DRAM core cell



3-Transistor DRAM operation

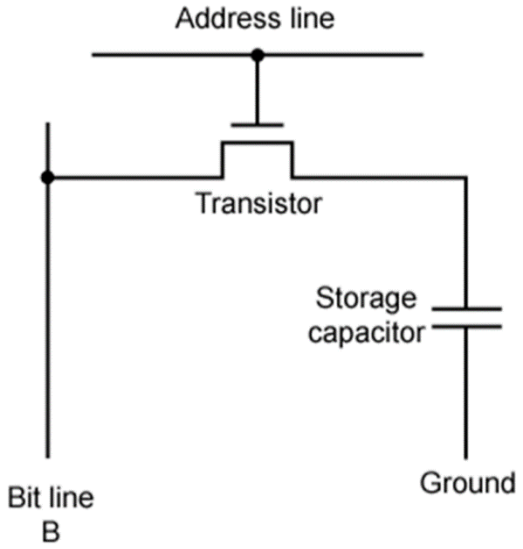
- Value is stored on gate capacitance of t_1.

- **Read:**
  - read = 1, write = 0, read_data' is precharged;
  - t_1 will pull down read_data' if 1 is stored.

- **Write:**
  - read = 0, write = 1, write_data = value;
  - guard transistor writes value onto gate capacitance.

1-Transistor DRAM Structure & Operation

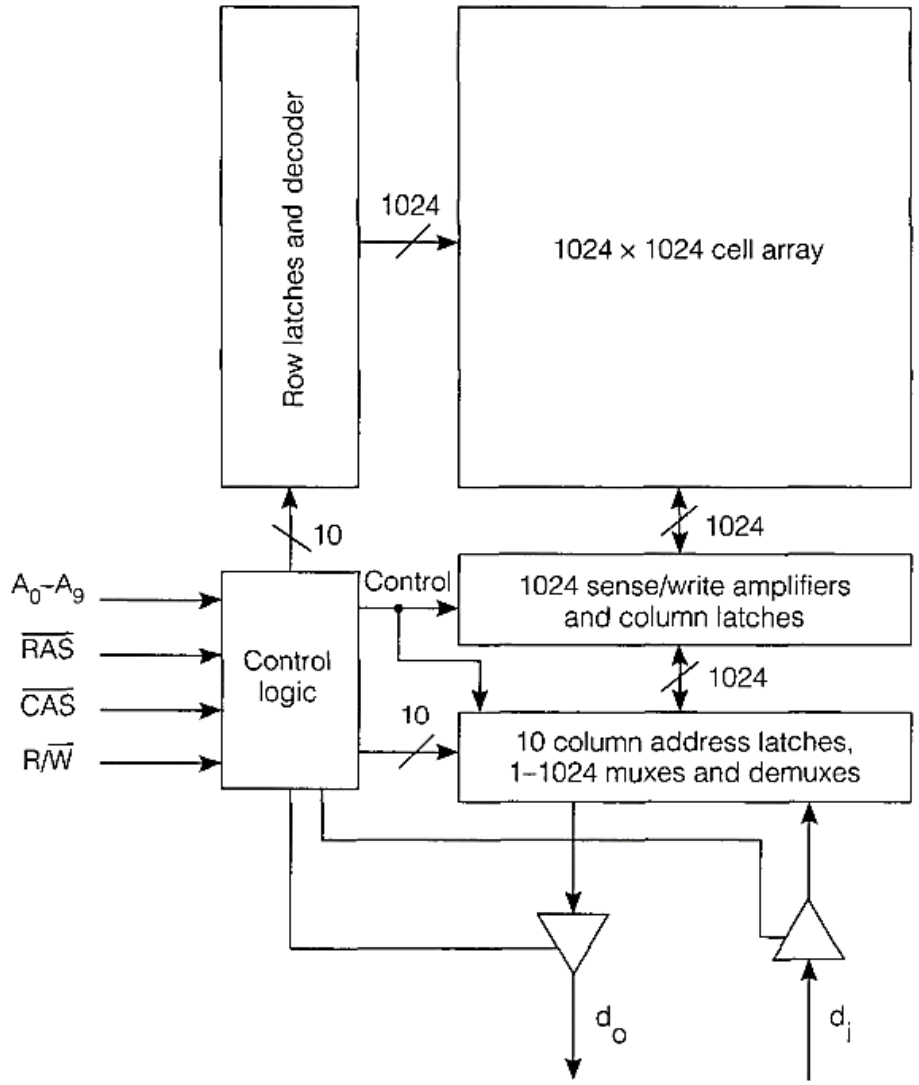
- Address line active when bit read or written
  - Transistor switch closed (current flows)
- Write
  - Voltage to bit line
    - High for 1 low for 0
  - Then signal address line
    - Transfers charge to capacitor
- Read
  - Address line selected
    - transistor turns on
  - Charge from capacitor fed via bit line to sense amplifier
    - Compares with reference value to determine 0 or 1
  - Capacitor charge must be restored



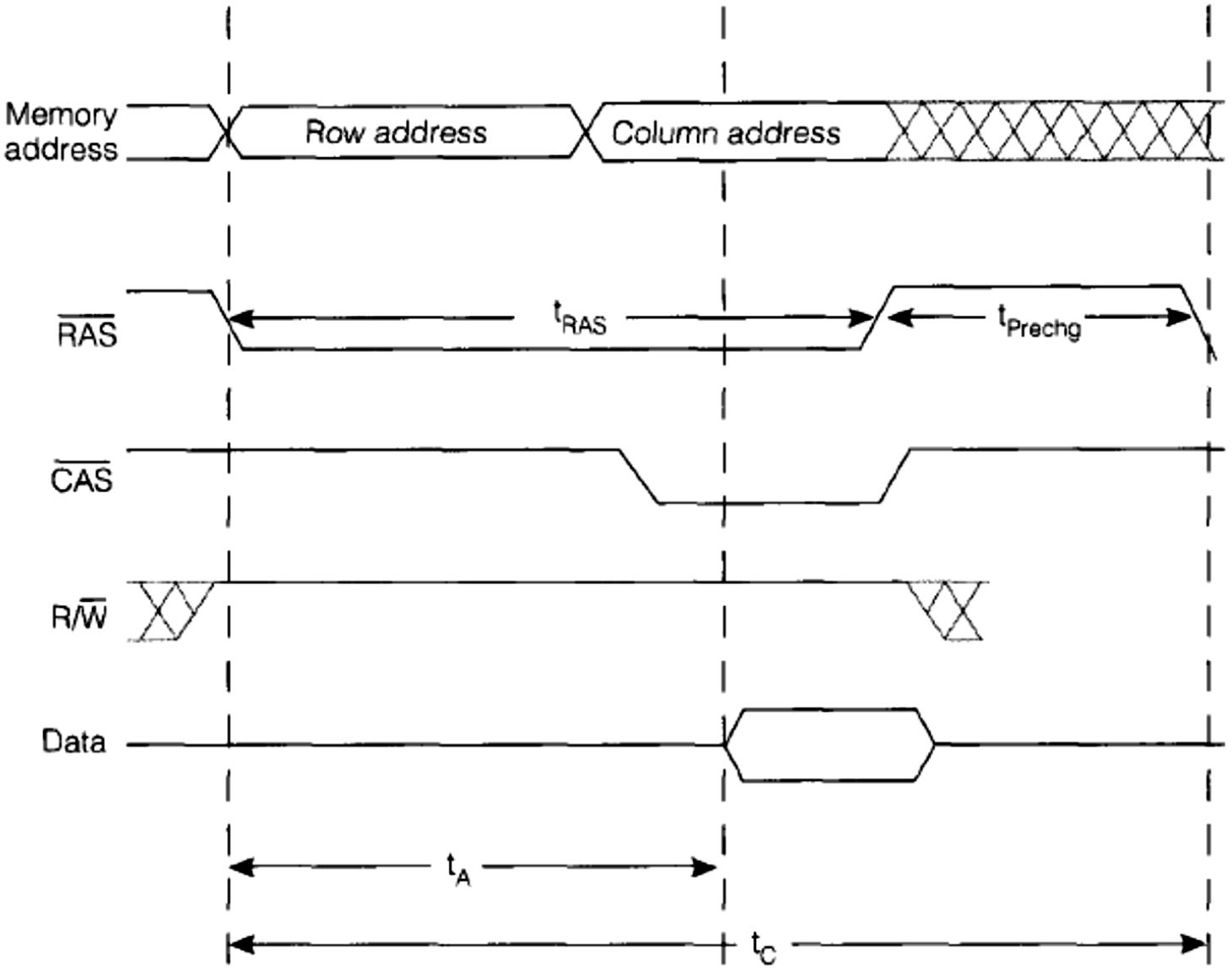
DRAM Refreshing

- Refresh circuit included on chip
  - Disable memory array chip
  - Count through rows and select each in turn
  - Read contents & write it back (restore)
- Takes time
- Slows down apparent performance

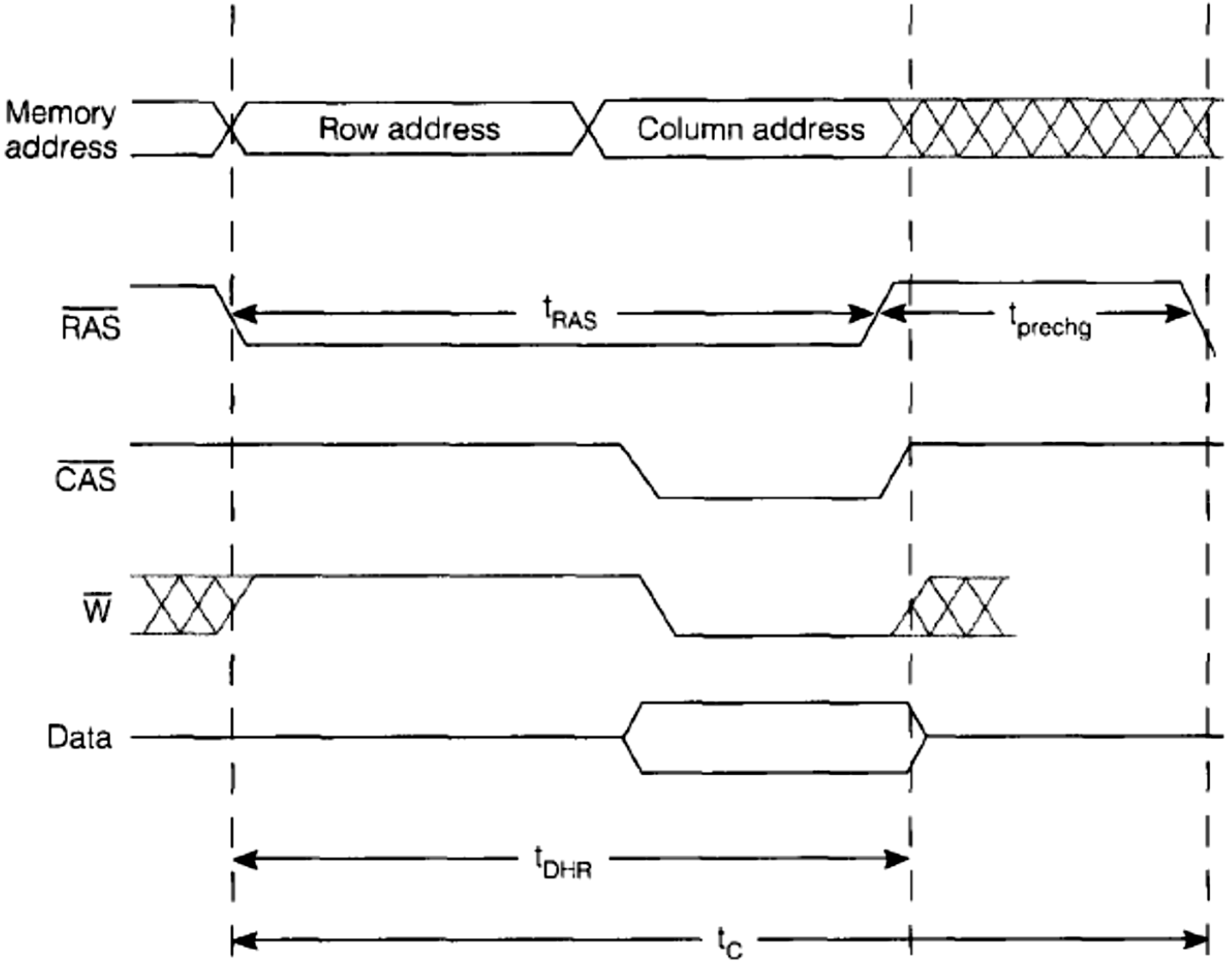
DRAM Chip organization



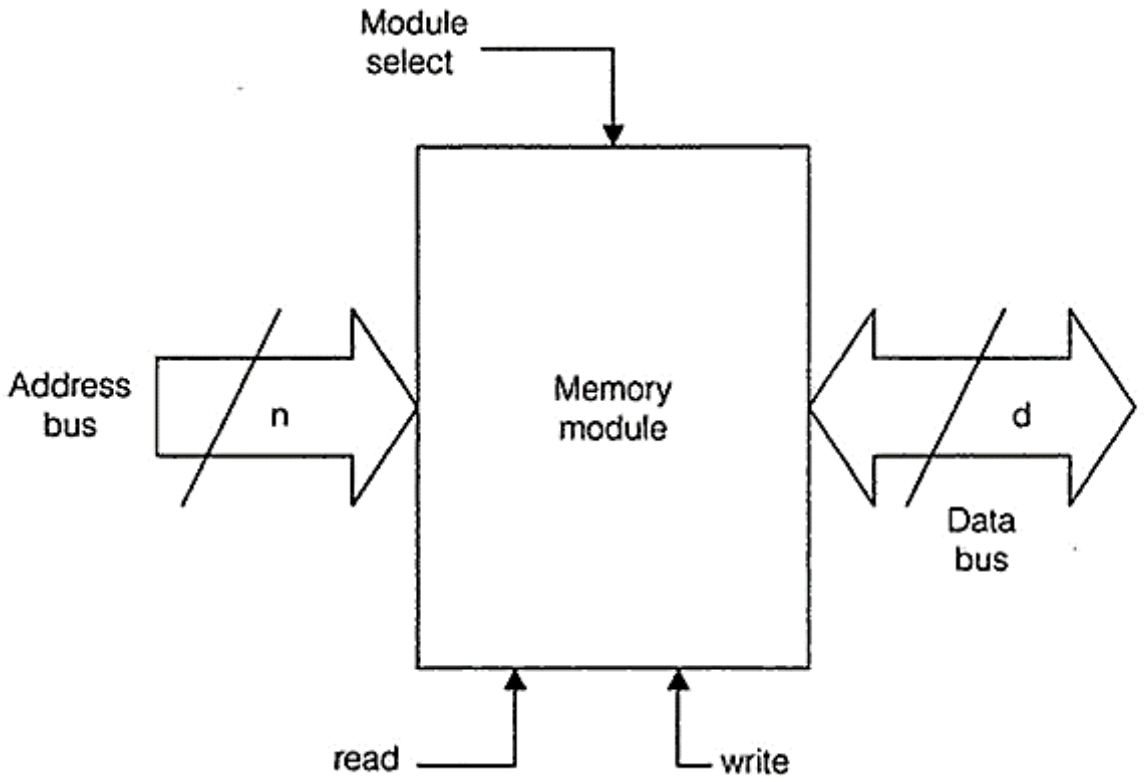
DRAM Timing: Read operation



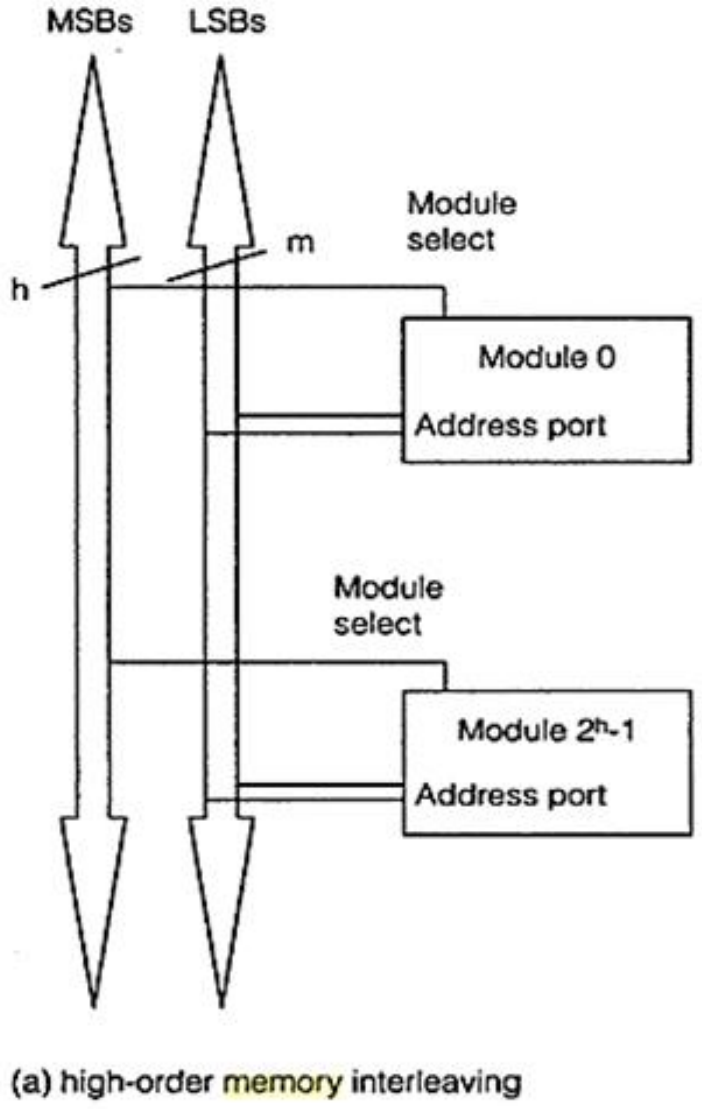
DRAM Timing: Write operation



Block diagram of a Memory Module



High order memory interleaving scheme

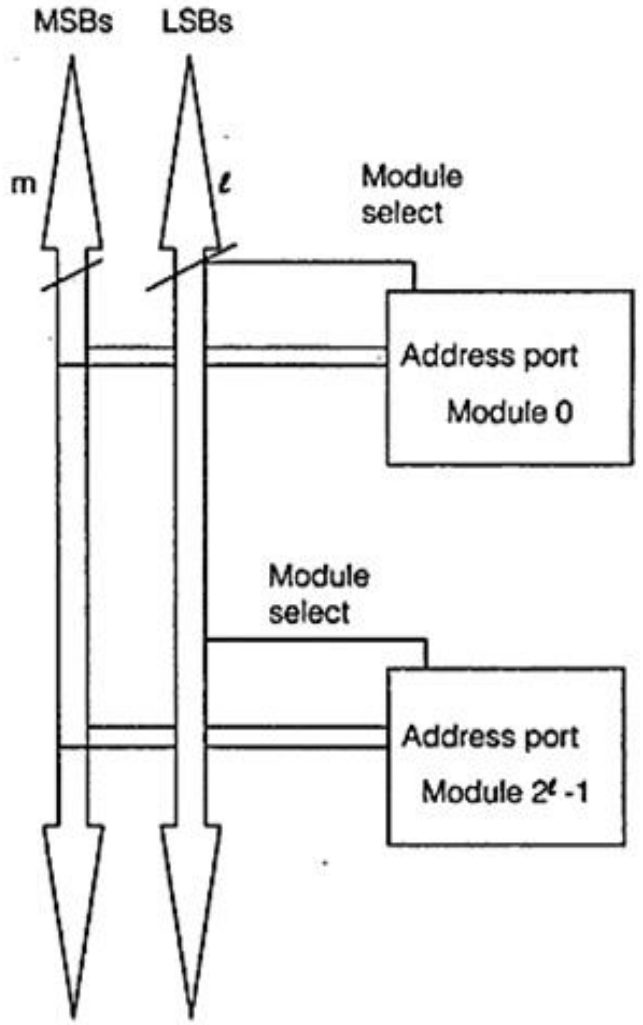


(a) high-order memory interleaving

- Total : $h+m$ address lines.
- h higher order to select modules (2^h modules)
- m lower order lines to address port
- Module size is 2^m -1.
- Consecutive addresses get mapped onto the same module.
- Addresses that gets mapped on to i th module is

$$i.M+j; \qquad 0 \leq i \leq 2^h-1 \text{ and } 0 \leq j \leq 2^m-1$$

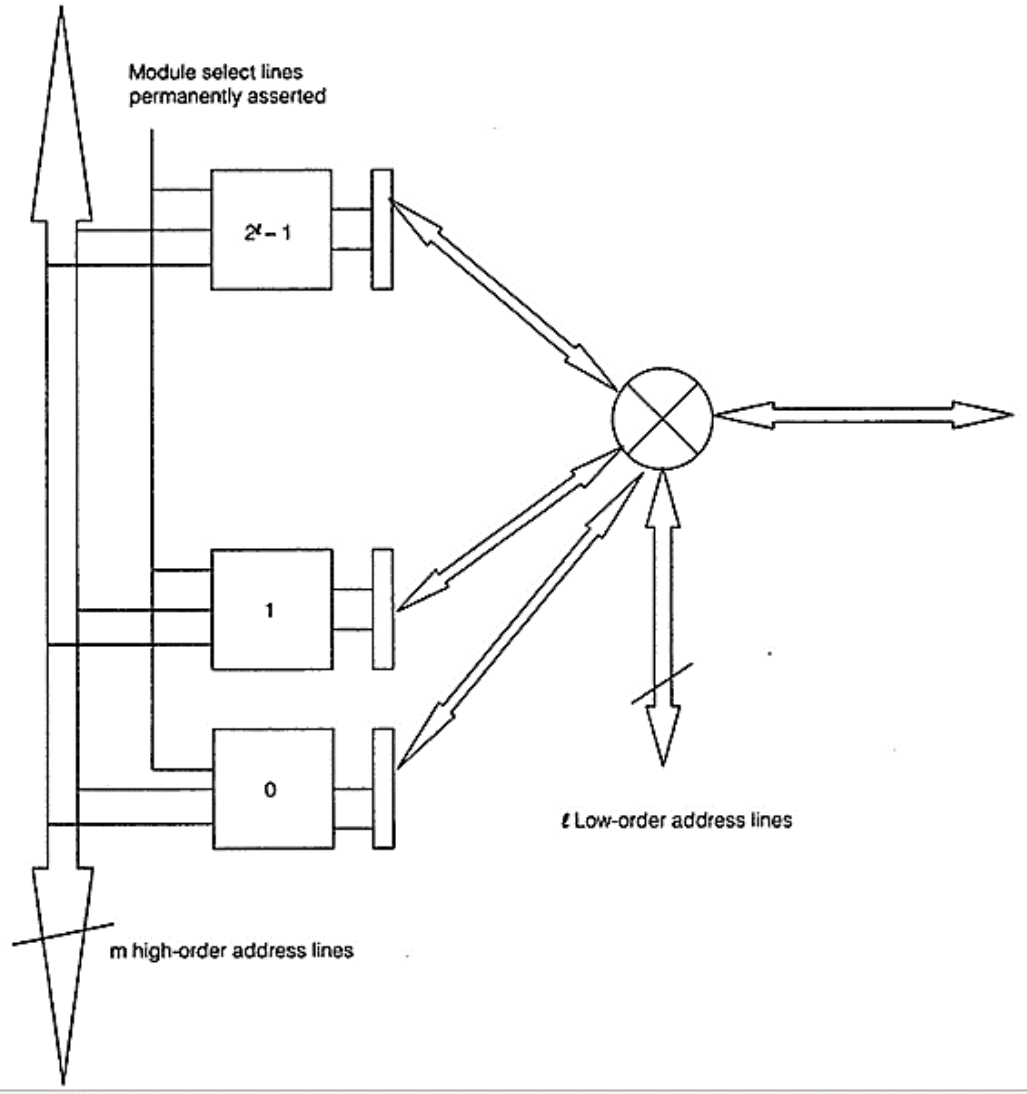
Low-order memory interleaving scheme



(b) low-order memory interleaving

- Total : $m+l$ address lines
- l lower-address lines for module selection,
- m higher order address to the address ports.
- Successive addresses map to successive modules
- addresses that gets mapped on to i th module is given by
  - $i+Nj; \quad 0 \leq i \leq 2^l-1$ and $0 \leq j \leq 2^m-1$

Simplex Low-order Memory Interleaving



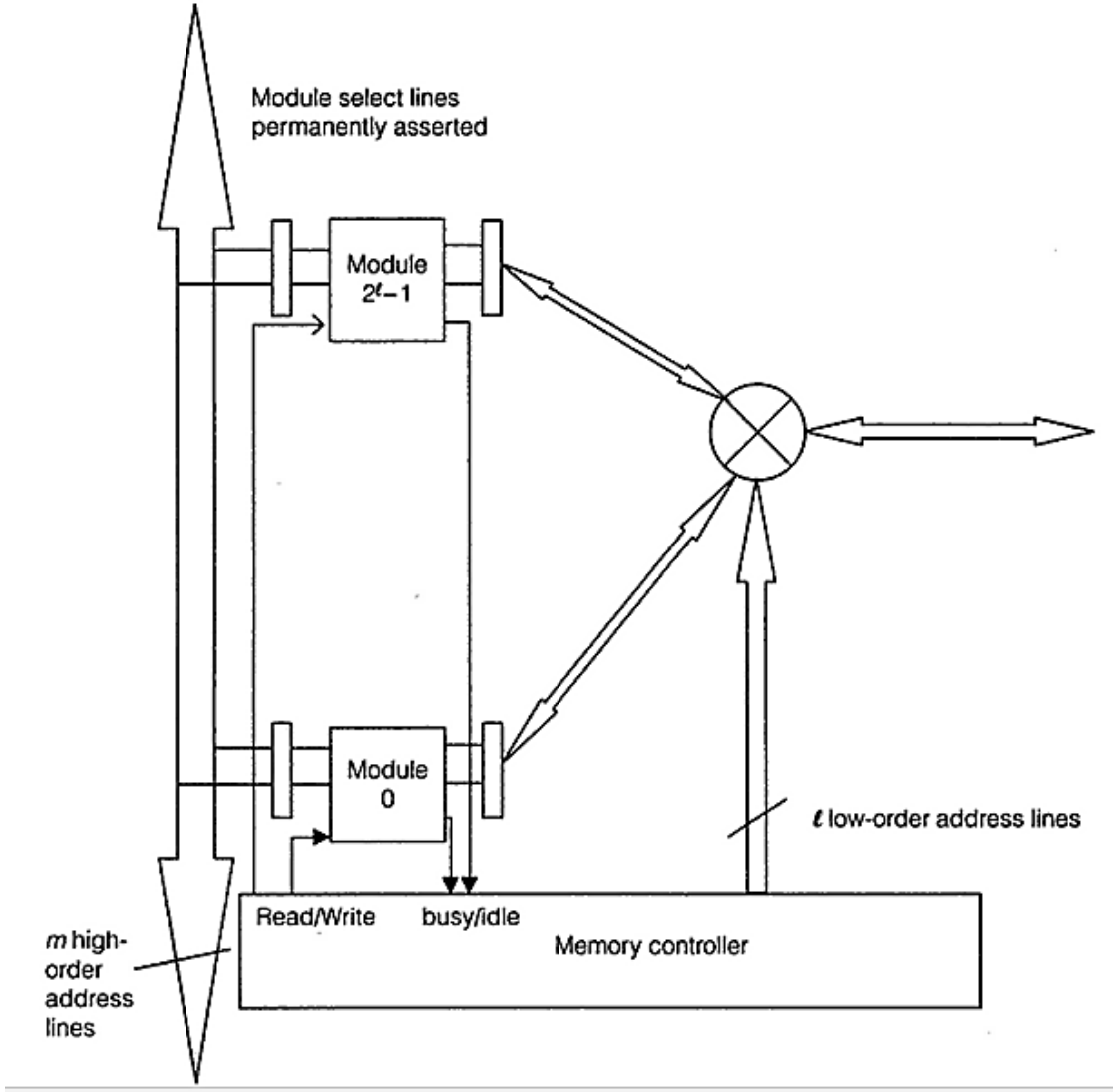
- T- access time of a module
- M – no of modules (= 2^{l})
-
- Effective memory access time for fetching data stored in successive addresses is T/ M.
- Access with skip distance d.
- T_{eff} = (T/N)*s for s < N
- T_{eff} = T for s ≥ N

Low – order interleaving: Address mapping.

7		7	15	23	31
6		6	14	22	30
5		5	13	21	29
4		4	12	20	28
3		3	11	19	27
2		2	10	18	26
1		1	9	17	25
0		0	8	16	24

- For d = 2, and starting address is even then
- access pattern is 0, 2, 4, 6, 8, 10, 12 , 14…
- Modules selected are 0, 2, 4, 6, 0, 2, 4, 6 …
- Only four modules can be accessed in parallel.
- Effective access time = 2 * T/M in simplex scheme.
- For l = 3, M =8 and skip distance d =3,
- Access pattern is 0, 3, 6, 9, 12, 18, 21.
- Modules accessed : 0, 3, 6, 1, 4, 7, 2, 5.
- Seq of 8 accesses map to 8 different modules.- Potential for parallel access.
- However high order lines are same only for (0, 3, 6) and different for (9, 12, 15).

Complex Low-order Memory Interleaving



- Latches before address port as well as data port
- Memory controller maps the address to the module.
- If module is free the high order address is latched in the address latch of the module.
- READ command is given. Module becomes busy.
- When module becomes idle controller routes the data to common data bus.
- T_{eff} = (T/N) if N and s are relative prime
- In general, T_{eff} = (T/N) times product of common prime factors of N and s

The Need for a Memory Hierarchy

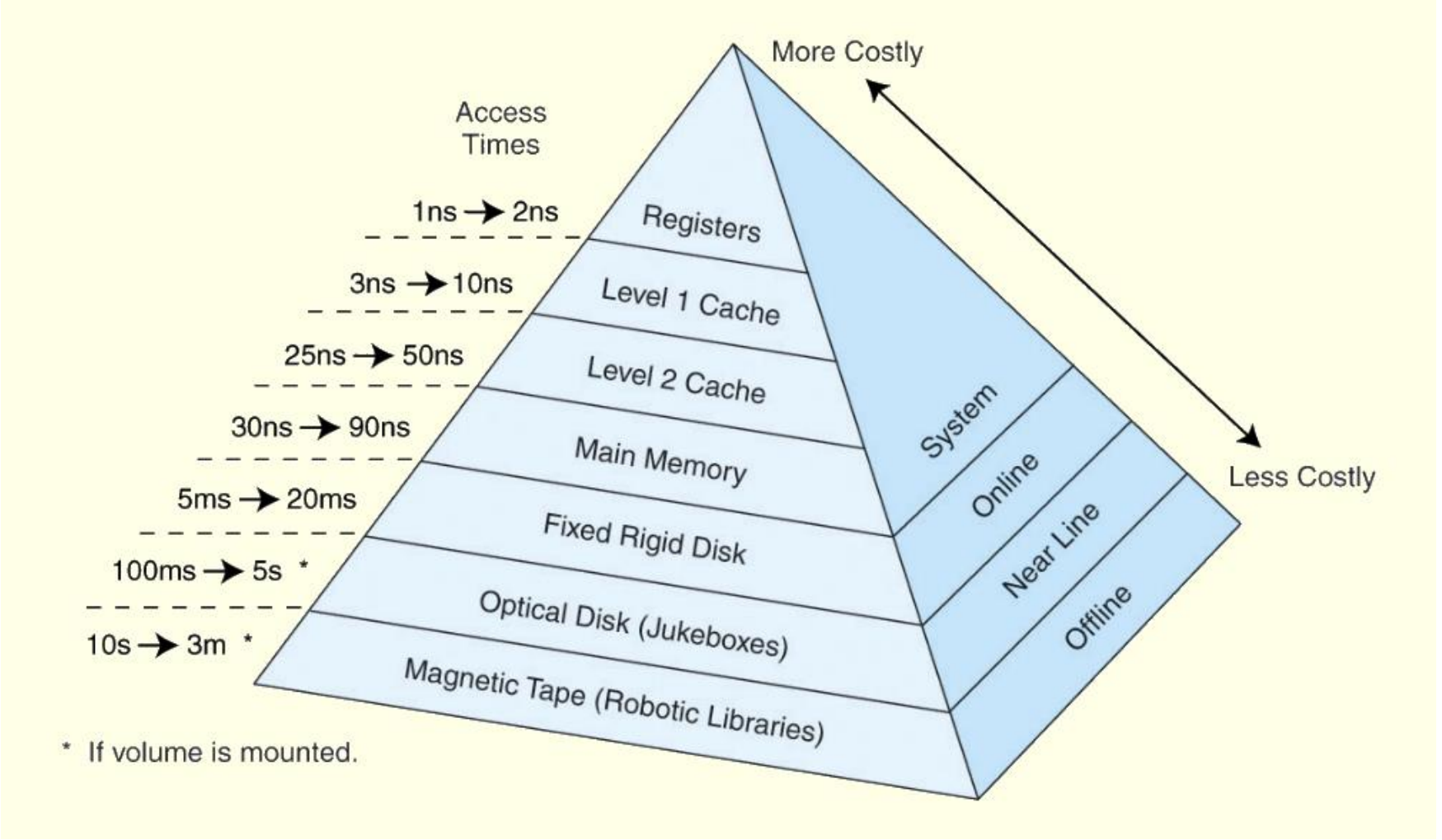
The widening speed gap between CPU and main memory

- Processor operations take of the order of 1 ns
- Memory access requires 10s or even 100s of ns

Memory bandwidth limits the instruction execution rate

- Each instruction executed involves at least one memory access hence, a few to 100s of MIPS is the best that can be achieved
- A fast buffer memory can help bridge the CPU-memory gap
- The fastest memories are expensive and thus not very large
- A second (third?) intermediate cache level is thus often used

Memory Hierarchy - Diagram



Memory Hierarchy: characteristics

- Let n is number memory levels.
- At i th memory level where 1<=i<=n,
- c_i – cost per bit
- t_i – average access time in seconds
- s_i – size of memory in bits
- b_i – memory access bandwidth in bits/seconds
- g_i – granularity of transfer from level i to i-1.
- $c_i > c_i+1$, (costly memory close to processor)
- $t_i < t_i+1$, (high speed memory close to processor)
- $s_i < s_i+1$, (smaller memory close to processor)
- $b_i > b_i+1$ and
- $g_i < g_i+1$

Memory Hierarchy: properties

Inclusion property:

- Address space in level i is a subset of that in level i+1.
- All data in level i should also exist in level i+1
- Doesn't necessarily mean data stored in address A in level i is same as that of what is stored in A in level i+1.

Coherence property:

Different copies of same data present at various levels of hierarchy should be identical (data consistency or coherency).

Memory hierarchy design problem

- h_i - hit ratio
- $(1-h_i)$ - miss ratio
- f_i - access frequency of level i

$$f_i = (1 - h_1)(1 - h_2) \ldots (1 - h_{i-1})h_i$$

- T_i – effective access time

$$T_i = \sum_{k=1}^{i} t_k$$

- Effective size = s_n (from inclusion property)
- T – effective access time of the n-level memory hierarchy

$$T = \sum_{i=1}^{n} f_i T_i$$

- C – total cost of memory hierarchy

$$C = \sum_{i=1}^{n} c_i s_i$$

- Large size, reasonable cost, low effective access time.

THANK YOU